

GUIDELINES IN WRITING SECTIONS V–VII OF THE SAMP

SECTION V: LOGGING, MONITORING, AND INCIDENT RESPONSE

This section explains how your system will detect security threats, monitor performance, and respond to incidents. Use FUTURE TENSE and relate your discussion to your study.

5.1 System Logging

- Describe the types of logs generated such as user logins, authentication attempts, system access, file changes, database transactions, network traffic, and security events.
- Explain the importance of logging for security monitoring, compliance, accountability, and digital forensics.
- Discuss log storage, retention, and protection from unauthorized modification.
- Write 2–3 technical paragraphs aligned with your study.

5.2 Monitoring and Alerts

- Explain the monitoring tools that will be used.
- Discuss alert triggers including failed logins, abnormal resource usage, suspicious activities, and service downtime.
- Describe health metrics such as CPU, memory, disk, uptime, application performance, and network usage.
- Provide a realistic and proactive monitoring strategy.

5.3 Incident Response

- Discuss detection methods using logs, alerts, and monitoring tools.
- Explain containment strategies such as isolating compromised systems and blocking attackers.
- Describe recovery including system restoration, backup recovery, and verification of integrity.
- Explain reporting including documentation, root cause analysis, and continuous improvement aligned with ISO/IEC 27035 and NIST.

5.4 Linux Monitoring and Log Analysis (MANDATORY)

You must perform hands-on monitoring and include the following:

- Run: `sudo tail -f /var/log/auth.log` and include screenshots.
- Explain how logs helped detect the incident and how the administrator responded.
- Provide a technical explanation of how logs help detect cyberattacks.

SECTION VI: PATCHING, UPDATES, AND VULNERABILITY MANAGEMENT

6.1 Update Policy

- Explain update schedules, automated and manual patching, emergency updates, and maintenance windows.
- Discuss testing patches before deployment and ensuring backups before updates.

6.2 Vulnerability Scanning

- Discuss regular vulnerability scanning and prioritization of risks.
- Explain remediation strategies and risk-based security.

6.3 Security Hardening

- Explain firewall configuration, secure authentication, encryption, and system hardening.
- Discuss disabling unnecessary services and secure system configuration.

6.1 Linux Security Updates (MANDATORY)

- Run: `sudo apt update` and `sudo apt upgrade`.
`sudo apt install lynis`
`sudo lynis audit system`
`sudo lynis audit system > lynis-report.txt`
- Include screenshots.
- Explain the importance of patching.